MAHATMA GANDHI UNIVERSITY

MGU-UGP (HONOURS) Regular EXAMINATION October 2025- Third SEMESTER

MG3VACBOT200 - BIOETHICS AND IPR (2024 ADMISSION ONWARDS)

\

Part A -Answer any 10 questions [1x10 = 10]

1. State one reason why the exchange of biological material across borders needs to be regulated. [K] / [CO1]
   To prevent the misuse or unauthorized exploitation of genetic resources T o ensure fair sharing of benefits arising from their use.
2. Write an example of a cultural issue in bioethics. [U] / [CO1)
   Organ donation and transplanttaion , use of human embryos in stem cell research, Human cloning (Any one or appropriate example)
3. Define informed consent. [U] / [CO1]
   Process of obtaining a person's voluntary agreement to participate in a medical procedure or research after being fully informed about its purpose, methods, risks, benefits etc.
4. Justify why ethical frameworks are essential in the utilization of biological material. [U] / [CO1]
   To protect the environment, to promote fairness and transparency and to prevent misuse or exploitation of biological resources
5. Which intellectual property right protects confidential business information? Trade secret [K] / [CO2]
6. In which year did India adopted National IPR Policy? | [K] / [CO3]
   2016
7. What is a plant variety? [K] / [CO4, CO5]
   A group of plants within a species that uniform characteristics such as appearance, yield, or resistance to pests and diseases, and can be reproduced or propagated consistently .
8. Define IPR [U] / [CO4, CO5]
   Legal rights that protect creations of the mind , such as inventions, literary and artistic works, designs, symbols, and brand names, giving the creator or owner exclusive rights to use and benefit from their work
9. What are the three main types of patents? [K] / [CO3, CO4]
   Utility, Design and Plant patents
10. What do you mean by patent infringement? [U] / [CO4, CO5]
    Unauthorized making, using, selling, or importing of a patented invention without the patent holder's permission
11. Name one international treaty relating to protection of TK. [K] / [CO4]
    Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge
12. What is the primary function of copyright law? [U] / [CO4]
    To encourage the creation of new works by granting creators exclusive rights over their original work for a limited time

Part B-Short Essay
Answer any 6 questions-[5x6 = 30]

13. Briefly explain the effect of GMOs on soil microorganisms. [U] / [CO1]

- GM plants e ngineered for herbicide tolerance, can indirectly benefit soil health by promoting no-till farming practices, which reduce erosion and maintain soil structure.
- GM crops do not adversely influence soil microbiological processes .
- GM crops may impact nutrient cycling in the rhizosphere soil.

14. Give an account on ethical concerns in professional streams. [U] / [CO1]

Honesty and integrity, Confidentiality, Conflict of interest, Accountability, Respect for human and animal rights, Environmental responsibility **(Short description).**

15. Explain the concept of novelty in patenting biotech inventions. [U] / [CO1]

The invention must be new and not previously known or disclosed anywhere in the world before the patent application is filed. The concept of novelty is one of the key requirements for granting a patent in biotechnology and other fields. It ensures that only truly new and original inventions receive protection. In biotechnology, novelty means that the invention—such as a new gene sequence, microorganism, genetically modified plant, or process— has not been made public anywhere in the world before the date of the patent application.

16. Explain the importance of GI. [U] / [CO4, CO5]
Protects regional identity, Prevents misuse, Promotes economic growth, Encourages rural development, Supports exports **(very short description)**

17. Write a note on the TRIPS Agreement and its significance. [U] / [CO3]

The TRIPS Agreement (Trade-Related Aspects of Intellectual Property Rights) is an international legal agreement administered by the World Trade Organization (WTO) . It was established in 1995 to set minimum standards for protecting and enforcing intellectual property rights (IPRs) among member countries.

18. Differentiate between a trademark and a patent. [U] / [CO2]

	Trademark	**Patent**
Definition	A sign, symbol, word, or logo that identifies and distinguishes the goods or services of one business from another.	A legal right granted for a new invention or process that provides a new way of doing something or offers a new technical solution.
Purpose	Protects brand identity and prevents misuse of business names or symbols.	Protects innovative ideas, inventions, and technologies.

	Trademark	Patent
Example	The Nike "swoosh" logo or Coca-Cola's brand name.	A new drug formula, a machine, or a genetically modified plant.

19. Define trademark infringement and explain its essential features. [U] / [CO5]

Trademark infringement is the unauthorized use of a registered trademark or a deceptively similar mark by another party, in a way that is likely to cause confusion, deception, or misunderstanding among consumers about the origin of goods or services.

- Unauthorized use: The trademark is used without the permission of the registered owner.
- Identical or deceptively similar mark: The infringing mark closely resembles the registered one in appearance, sound, or meaning.

20. Discuss the challenges in implementing data protection laws. [E] / [CO4, CO5]
Lack of awareness, Rapid technological changes, Cross-border data flow, Weak enforcement mechanisms, Cybersecurity threats (Brief description)

Part C- Answer any one [10x1 = 10]

21. Describe the main ethical concerns related to prenatal diagnosis, gene therapy, and organ transplantation. [U] / [CO1]

1. Ethical Concerns in Prenatal Diagnosis

- Risk of selective abortion: It may lead to ending pregnancies based on the baby's sex or genetic conditions, raising moral questions.
- Discrimination: Can increase negative attitudes toward people with disabilities or genetic disorders.
- Privacy issues: Genetic information of the unborn child may be misused or disclosed without consent.
- Emotional and psychological stress: Parents may face guilt, fear, or pressure when making decisions after receiving test results.
- Religious and moral beliefs: Some people believe that interfering with natural birth processes is morally or religiously wrong.

2. Ethical Concerns in Gene Therapy

- Safety and uncertainty: Gene therapy may cause unintended effects or new genetic problems in future generations.
- Fair access: The treatment may be expensive and available only to the wealthy, raising social inequality concerns.
- Genetic enhancement: Using gene therapy to improve traits like intelligence or looks creates ethical debates.
- Informed consent: Patients must fully understand the risks before undergoing experimental treatments.

- Impact on future generations: Changes in genes could permanently alter the human gene pool, which may have unpredictable results.

3. Ethical Concerns in Organ Transplantation

- Organ trade and exploitation: Poor people may be exploited for their organs due to financial need.
- Fair distribution: Ethical issues arise over how organs are allocated—who gets priority and why.
- Consent and autonomy: Organs should only be taken with proper consent from donors or their families.
- Cultural and religious beliefs: Some traditions may object to organ removal after death.
- Use of animal or artificial organs: Raises questions about crossing species boundaries and long-term safety.

22. Evaluate the role of trademarks in branding and consumer protection. [E] / [CO4, CO5]

1. Role in Branding

- Identity and recognition: A trademark gives a unique identity to a product or company, helping consumers easily recognize it among competitors.
- Brand loyalty: When customers trust a brand's quality, its trademark helps build loyalty and repeat purchases.
- Marketing tool: Trademarks make advertising more effective, as logos, names, or symbols attract attention and convey brand values.
- Business reputation: A strong trademark reflects a company's reputation and goodwill in the market.
- Competitive advantage: Trademarks legally protect brand elements, preventing others from copying or imitating them.

2. Role in Consumer Protection

- Guarantee of quality: A registered trademark assures consumers that the product comes from a trusted source with consistent standards.
- Prevents confusion: It helps consumers distinguish between genuine and fake or inferior products.
- Legal protection: Trademarks protect buyers from misleading or deceptive marketing practices.
- Accountability: Consumers can identify and hold the manufacturer responsible for defective or harmful goods.
- Encourages fair trade: By preventing imitation and fraud, trademarks promote honesty and transparency in the market. Top of Form Bottom of Form